

Lab 18: Investigating Pertussis Resurgence

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Background

Pertussis (more commonly known as whooping cough) is a highly contagious respiratory disease caused by the bacterium *Bordetella pertussis*.

We will be using web-scraping, JSON based APIs and advanced dplyr and ggplot to investigate brand new datasets associated with pertussis cases and longitudinal RNA-Seq on the immune response to distinct vaccination strategies.

Investigating pertussis cases by year

The United States Centers for Disease Control and Prevention (CDC) has been compiling reported pertussis case numbers since 1922 in their National Notifiable Diseases Surveillance System (NNDSS). We can view this data on the CDC website here: <https://www.cdc.gov/pertussis/surv-reporting/cases-by-year.html>

```
read.csv("Pertussis Cases 1922-2025.csv")
```

	Year	Number.of.Reported.Pertussis.Cases	Data.Status
1	1922	107,473	Finalized
2	1923	164,191	Finalized
3	1924	165,418	Finalized
4	1925	152,003	Finalized
5	1926	202,210	Finalized
6	1927	181,411	Finalized
7	1928	161,799	Finalized
8	1929	197,371	Finalized
9	1930	166,914	Finalized
10	1931	172,559	Finalized
11	1932	215,343	Finalized
12	1933	179,135	Finalized
13	1934	265,269	Finalized
14	1935	180,518	Finalized
15	1936	147,237	Finalized
16	1937	214,652	Finalized
17	1938	227,319	Finalized
18	1939	103,188	Finalized
19	1940	183,866	Finalized
20	1941	222,202	Finalized
21	1942	191,383	Finalized
22	1943	191,890	Finalized
23	1944	109,873	Finalized
24	1945	133,792	Finalized
25	1946	109,860	Finalized
26	1947	156,517	Finalized
27	1948	74,715	Finalized
28	1949	69,479	Finalized
29	1950	120,718	Finalized
30	1951	68,687	Finalized
31	1952	45,030	Finalized
32	1953	37,129	Finalized
33	1954	60,886	Finalized
34	1955	62,786	Finalized
35	1956	31,732	Finalized
36	1957	28,295	Finalized
37	1958	32,148	Finalized
38	1959	40,005	Finalized
39	1960	14,809	Finalized
40	1961	11,468	Finalized
41	1962	17,749	Finalized
42	1963	17,135	Finalized

43	1964	13,005	Finalized
44	1965	6,799	Finalized
45	1966	7,717	Finalized
46	1967	9,718	Finalized
47	1968	4,810	Finalized
48	1969	3,285	Finalized
49	1970	4,249	Finalized
50	1971	3,036	Finalized
51	1972	3,287	Finalized
52	1973	1,759	Finalized
53	1974	2,402	Finalized
54	1975	1,738	Finalized
55	1976	1,010	Finalized
56	1977	2,177	Finalized
57	1978	2,063	Finalized
58	1979	1,623	Finalized
59	1980	1,730	Finalized
60	1981	1,248	Finalized
61	1982	1,895	Finalized
62	1983	2,463	Finalized
63	1984	2,276	Finalized
64	1985	3,589	Finalized
65	1986	4,195	Finalized
66	1987	2,823	Finalized
67	1988	3,450	Finalized
68	1989	4,157	Finalized
69	1990	4,570	Finalized
70	1991	2,719	Finalized
71	1992	4,083	Finalized
72	1993	6,586	Finalized
73	1994	4,617	Finalized
74	1995	5,137	Finalized
75	1996	7,796	Finalized
76	1997	6,564	Finalized
77	1998	7,405	Finalized
78	1999	7,298	Finalized
79	2000	7,867	Finalized
80	2001	7,580	Finalized
81	2002	9,771	Finalized
82	2003	11,647	Finalized
83	2004	25,827	Finalized
84	2005	25,616	Finalized
85	2006	15,632	Finalized

86	2007	10,454	Finalized
87	2008	13,278	Finalized
88	2009	16,858	Finalized
89	2010	27,550	Finalized
90	2011	18,719	Finalized
91	2012	48,277	Finalized
92	2013	28,639	Finalized
93	2014	32,971	Finalized
94	2015	20,762	Finalized
95	2016	17,972	Finalized
96	2017	18,975	Finalized
97	2018	15,609	Finalized
98	2019	18,617	Finalized
99	2020	6,124	Finalized
100	2021	2,116	Finalized
101	2022	3,044	Finalized
102	2023	7,063	Finalized
103	2024	43,321	Provisional
104	2025	28,783	Provisional

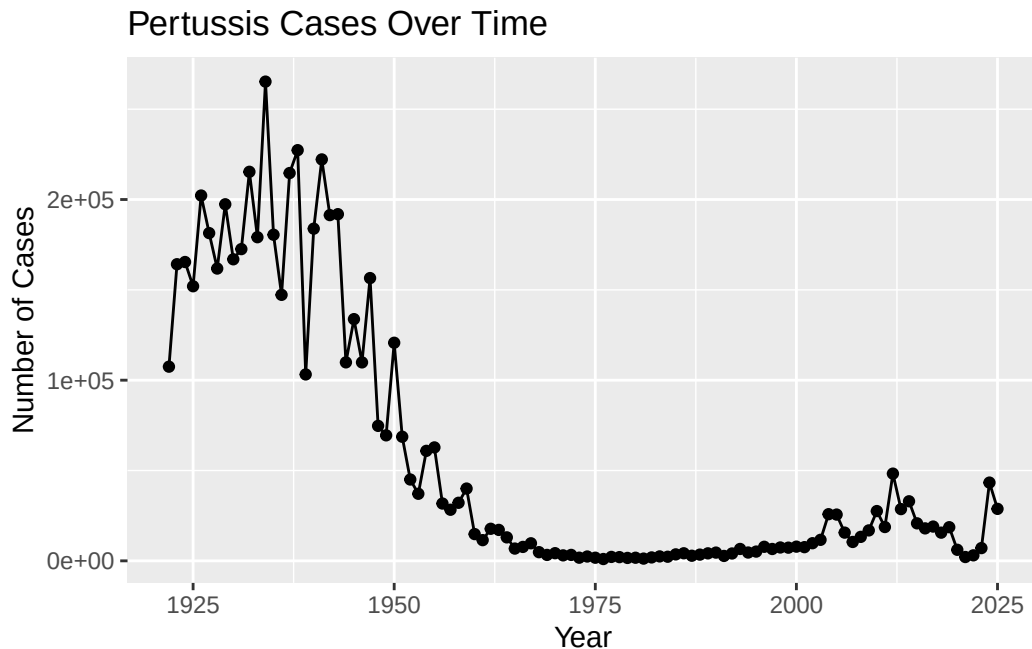
```
cdc <- read.csv("Pertussis Cases 1922-2025.csv")
```

```
cdc$Number.of.Reported.Pertussis.Cases <- as.numeric(gsub(",", "", cdc$Number.of.Reported.Pertussis.Cases))
```

Q1. Assign the CDC pertussis case number data to a data frame called `cdc` and use `ggplot` to make a plot of cases numbers over time.

```
library(ggplot2)
```

```
ggplot(cdc) +
  aes(x = Year, y = Number.of.Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  labs(
    x = "Year",
    y = "Number of Cases",
    title = "Pertussis Cases Over Time"
  )
```



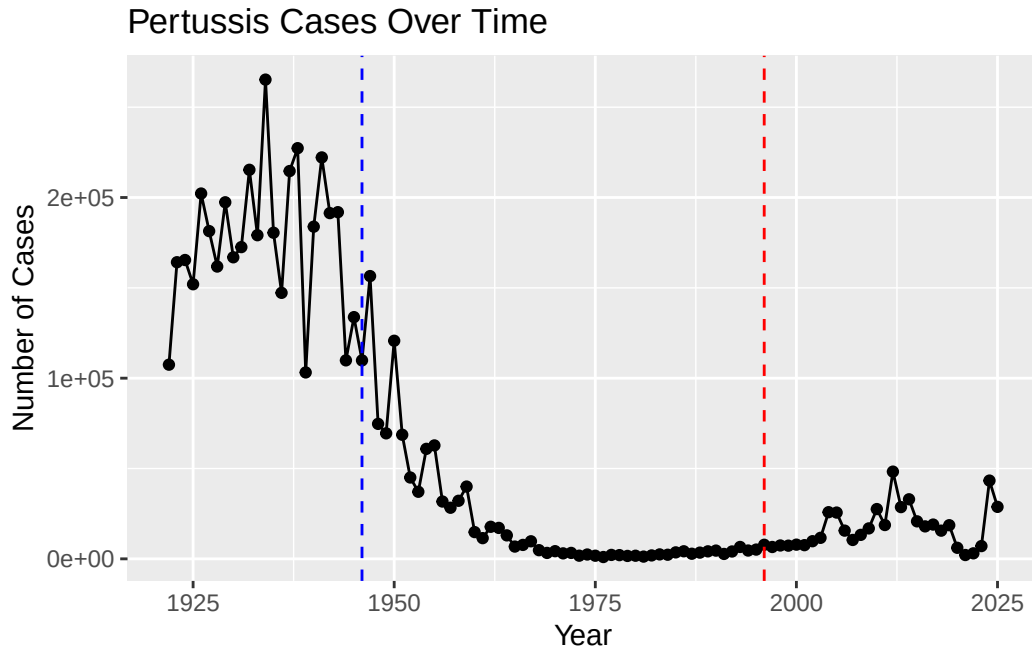
A tale of two vaccines (wP and aP)

Two types of pertussis vaccines have been developed: whole-cell pertussis (wP) and acellular pertussis (aP). The first vaccines were composed of ‘whole cell’ (wP) inactivated bacteria. The latter aP vaccines use purified antigens of the bacteria.

These aP vaccines were developed to have less side effects than the older wP vaccines and are now the only form administered in the United States.

Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine. What do you notice?

```
ggplot(cdc) +
  aes(x = Year, y = Number.of.Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1946, linetype = "dashed", color = "blue") +
  geom_vline(xintercept = 1996, linetype = "dashed", color = "red") +
  labs(x = "Year", y = "Number of Cases", title = "Pertussis Cases Over Time"
  )
```



Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the introduction of the aP vaccine (red dashed line) the number of cases appear to be rising once more. Some possible explanations for this trend could be due to the rise in hesitation of getting vaccinated, bacterial evolution, and difference in immunity to the vaccine in infants and young adults.

Exploring CMI-PB Data

Why is this vaccine-preventable disease on the upswing? To answer this question we need to investigate the mechanisms underlying waning protection against pertussis. This requires evaluation of pertussis-specific immune responses over time in wP and aP vaccinated individuals.

```
# Allows us to read, write and process JSON data
library(jsonlite)
```

Let's now read the main subject database table from the CMI-PB API:

```
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
```

Key-Point: The subject table provides metadata about each individual in the study group. For example, their infancy vaccination type, biological sex, year of birth, time of boost etc.

```
head(subject, 3)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
aP wP  
87 85
```

There are 87 aP infancy vaccinated subjects and 85 wP infancy vaccinated subjects in the dataset.

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```
Female    Male  
    112     60
```

There are 112 female and 60 male subjects/patients in the dataset.

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

Side Note: Working with Dates

Dates and times can be annoying to work with at the best of times. However, in R we have the excellent lubridate package, which can make life a lot easier.

```
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

Convert date into proper date format:

```
subject$year_of_birth <- ymd(subject$year_of_birth)
subject$date_of_boost <- ymd(subject$date_of_boost)
```

Create age in years:

```
subject$age <- time_length(interval(subject$year_of_birth, subject$date_of_boost), "years")
```

Q7. Determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

The average age of wP individuals:

```
mean(subject$age[subject$infancy_vac == "wP"], na.rm = TRUE)
```



```
[1] 30.3625
```

The average age of aP individuals:

```
mean(subject$age[subject$infancy_vac == "aP"], na.rm = TRUE)
```

```
[1] 21.91779
```

Are they significantly different? Let's perform a t-test that compares the mean ages of the two groups (wP / aP):

```
t.test(age ~ infancy_vac, data = subject)
```

Welch Two Sample t-test

```
data: age by infancy_vac
t = -13.31, df = 128.26, p-value < 2.2e-16
alternative hypothesis: true difference in means between group aP and group wP is not equal to 0
95 percent confidence interval:
 -9.700130 -7.189298
sample estimates:
mean in group aP mean in group wP
      21.91779      30.36250
```

A p-value of 2.2e-16 reveals that the average age between both groups IS statistically significant.

Q8. Determine the age of all individuals at time of boost?

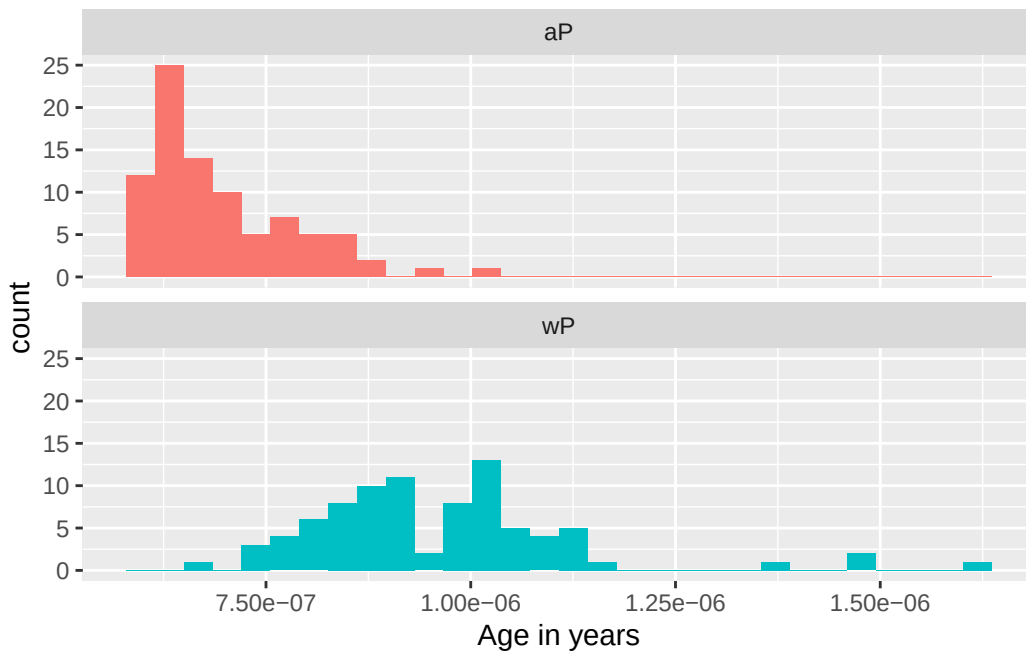
```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Yes, the histograms above clearly show that the two groups are significantly different. While the age for aP appears to be more skewed to the left, the age for wP is more skewed to the right. aP has more younger subjects whereas the subjects for wP are much older.

Joining Multiple Tables

We will read the specimen and ab_titer tables into R and store the data as specimen and titer named data frames:

```
specimen <- read_json("https://www.cmi-pb.org/api/specimen", simplifyVector = TRUE)
titer <- read_json("https://www.cmi-pb.org/api/plasma_ab_titer", simplifyVector = TRUE)
```

To know whether a given `specimen_id` comes from an aP or wP individual we need to link (a.k.a. “join” or merge) our specimen and subject data frames. The excellent dplyr package has a family of `join()` functions that can help us with this common task:

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
meta <- left_join(specimen, subject)
```

Joining with `by = join\_by(subject\_id)`

```
dim(meta)
```

```
[1] 1503  14
```

```
head(meta)
```

	specimen_id	subject_id	actual_day_relative_to_boost			
1	1	1	-3			
2	2	1	1			
3	3	1	3			
4	4	1	7			
5	5	1	11			
6	6	1	32			
	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex	

1	0	Blood	1	wP	Female
2	1	Blood	2	wP	Female
3	3	Blood	3	wP	Female
4	7	Blood	4	wP	Female
5	14	Blood	5	wP	Female
6	30	Blood	6	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

	age
1	30.69672
2	30.69672
3	30.69672
4	30.69672
5	30.69672
6	30.69672

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(titer, meta)
```

Joining with `by = join\_by(specimen\_id)`

```
dim(abdata)
```

```
[1] 52576    21
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)
```

IgE	IgG	IgG1	IgG2	IgG3	IgG4
6698	5389	10117	10124	10124	10124

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
      31520      8085      7301      5670
```

The different \$dataset values in abdata are as follows: 31520, 8085, 7301, and 5670. The most “recent” dataset shows a significant decrease compared to the value of rows under the older dataset.

Examine IgG Ab titer levels

Now using our joined/merged/linked abdata dataset we will `filter()` for IgG isotype:

```
igg <- abdata %>% filter(isotype == "IgG")
head(igg)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgG	TRUE	PT	68.56614	3.736992
2	1	IgG	TRUE	PRN	332.12718	2.602350
3	1	IgG	TRUE	FHA	1887.12263	34.050956
4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457

	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost
1	IU/ML	0.530000	1	-3
2	IU/ML	6.205949	1	-3
3	IU/ML	4.679535	1	-3
4	IU/ML	0.530000	3	-3
5	IU/ML	6.205949	3	-3
6	IU/ML	4.679535	3	-3

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	0	Blood	1	wP	Female
3	0	Blood	1	wP	Female
4	0	Blood	1	wP	Female
5	0	Blood	1	wP	Female
6	0	Blood	1	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

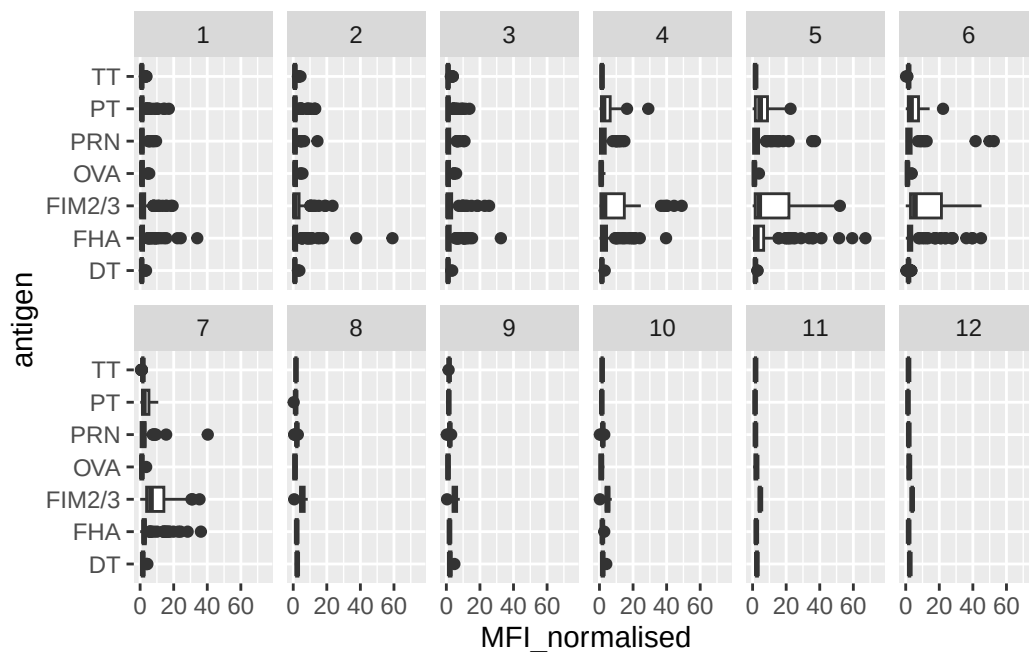
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
4		Unknown	White	1983-01-01	2016-10-10	2020_dataset
5		Unknown	White	1983-01-01	2016-10-10	2020_dataset
6		Unknown	White	1983-01-01	2016-10-10	2020_dataset

age

1	30.69672
2	30.69672
3	30.69672
4	33.77322
5	33.77322
6	33.77322

Q13. Make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```



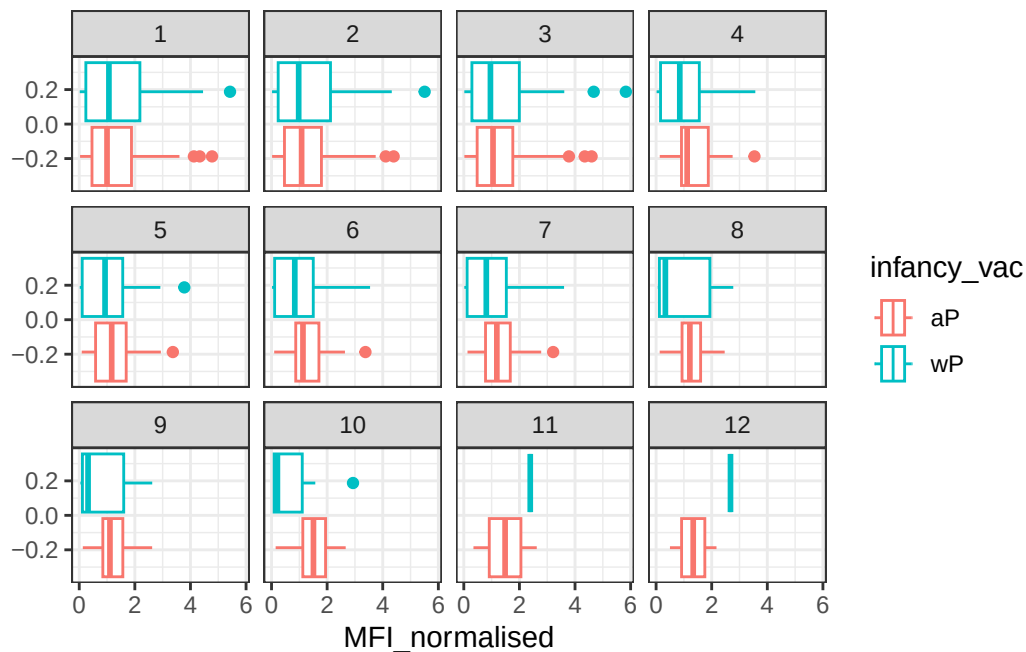
Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

The antigens that show differences in the level of IgG antibody titers recognizing them over time are FHA and FIM2/2. This makes sense as both of these antigens play crucial roles in bacteria adhesion, facilitating the adhesion of bacteria to host cells. The other antigens don't show much of a difference as they don't have a similar role to FHA and FIM2/2. For example, the antigen TT doesn't show a difference because it is a tumor associated antigen playing a role in immune response rather than bacteria adhesion.

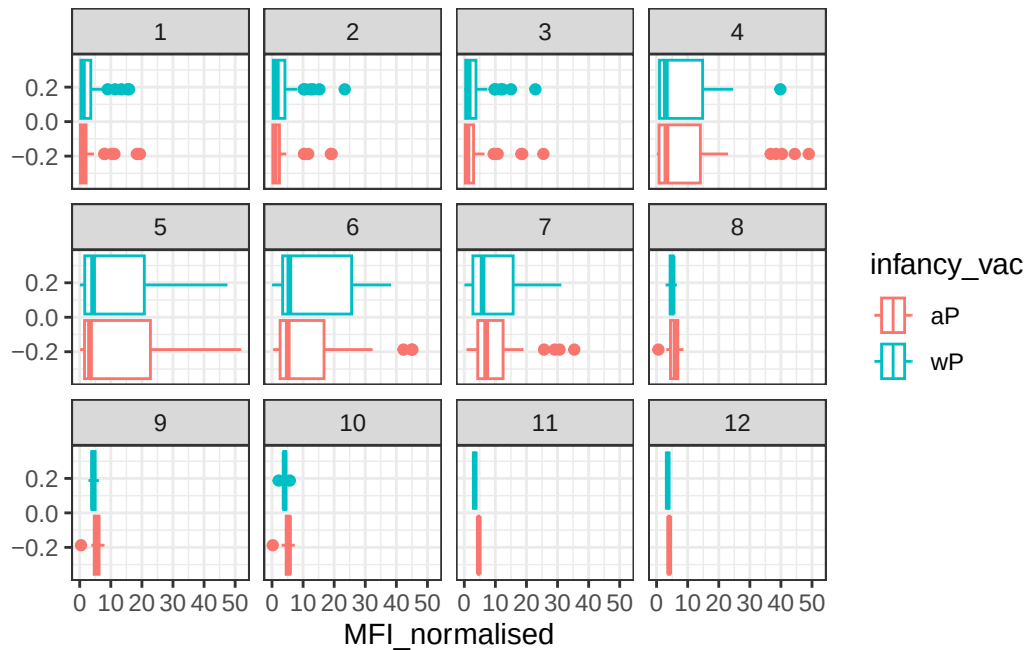
We can attempt to examine differences between wP and aP here by setting color and/or facet values of the plot to include infancy_vac status. However these plots tend to be rather busy and thus hard to interpret easily.

Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each.

```
filter(igg, antigen=="OVA") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = TRUE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



```
filter(igg, antigen=="FIM2/3") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = TRUE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



Q16. What do you notice about these two antigens time courses and the PT data in particular?

Whereas OVA remains stable FIM2/3 does not and drastically decreases over time. One thing to note about the PT data is that through plots 4-7 for FIM2/3 there is a clear stable increase before a sudden decrease, a pattern not found for OVA.

Q17. Do you see any clear difference in aP vs. wP responses?

aP and wP responses appear to be somewhat the same for FIM2/3 with aP appearing to be slightly quicker in visit 8. There is a clearer difference in OVA where aP shows a much quicker response than wP.

Lets finish this section by looking at the 2021 dataset IgG PT antigen levels time-course:


```

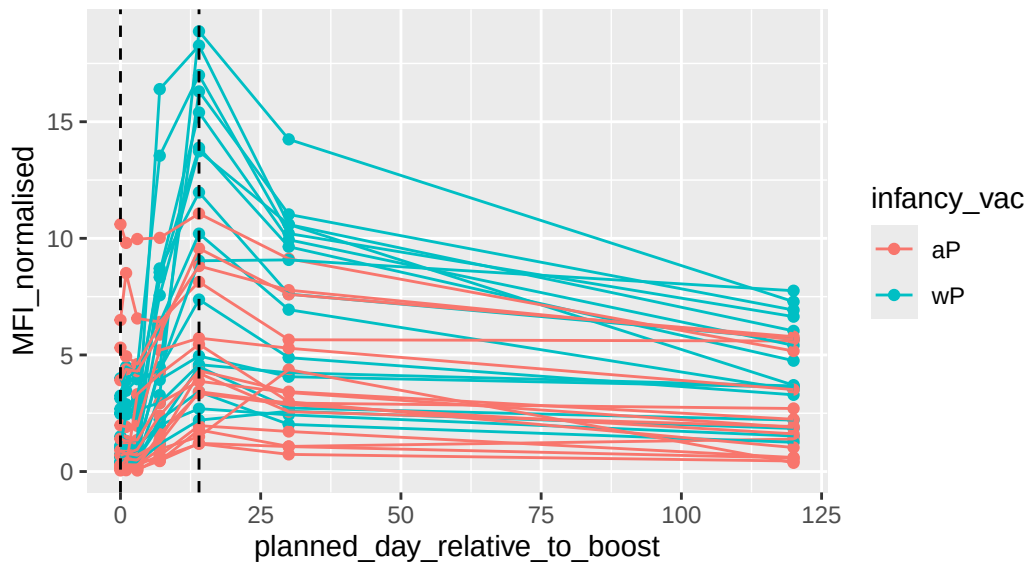
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2021 dataset IgG PT",
         subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")

```

2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



Q18. Does this trend look similar for the 2020 dataset?

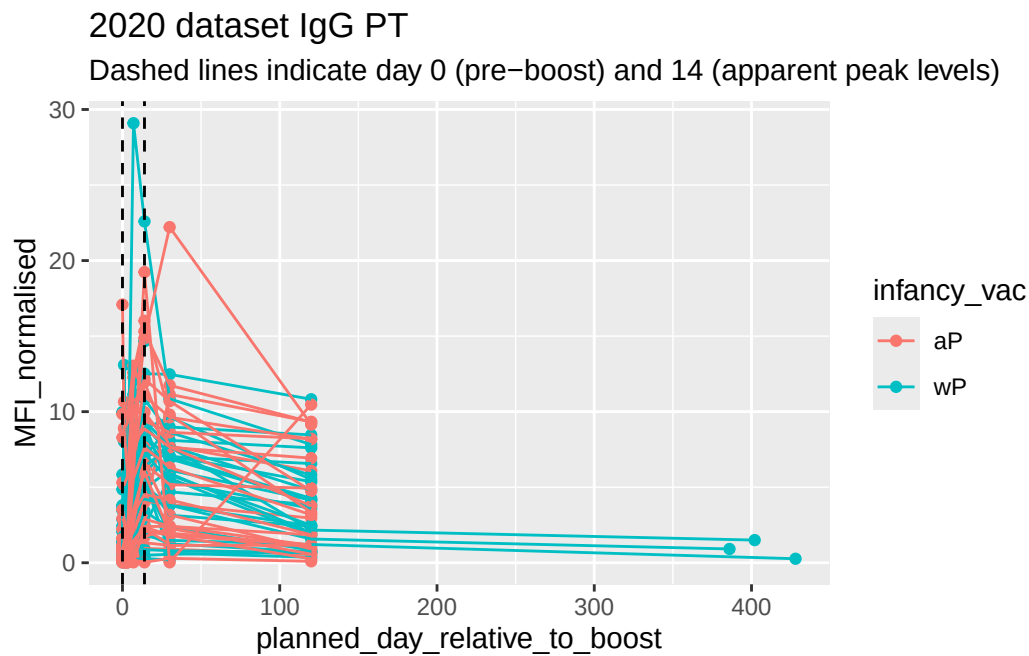
```

abdata.20 <- abdata %>% filter(dataset == "2020_dataset")

abdata.20 %>%
  filter(isotype == "IgG", antigen == "PT") %>%

```

```
ggplot() +
  aes(x=planned_day_relative_to_boost,
      y=MFI_normalised,
      col=infancy_vac,
      group=subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept=0, linetype="dashed") +
  geom_vline(xintercept=14, linetype="dashed") +
  labs(title="2020 dataset IgG PT",
       subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```



This trend does not appear to look similar for the 2020 dataset.

Obtaining CMI-PB RNASeq data

Let's read available RNA-Seq data for the gene IGHG1 into R and investigate the time course of its gene expression values.

```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSEG00000211896.7"
rna <- read_json(url, simplifyVector = TRUE)
```

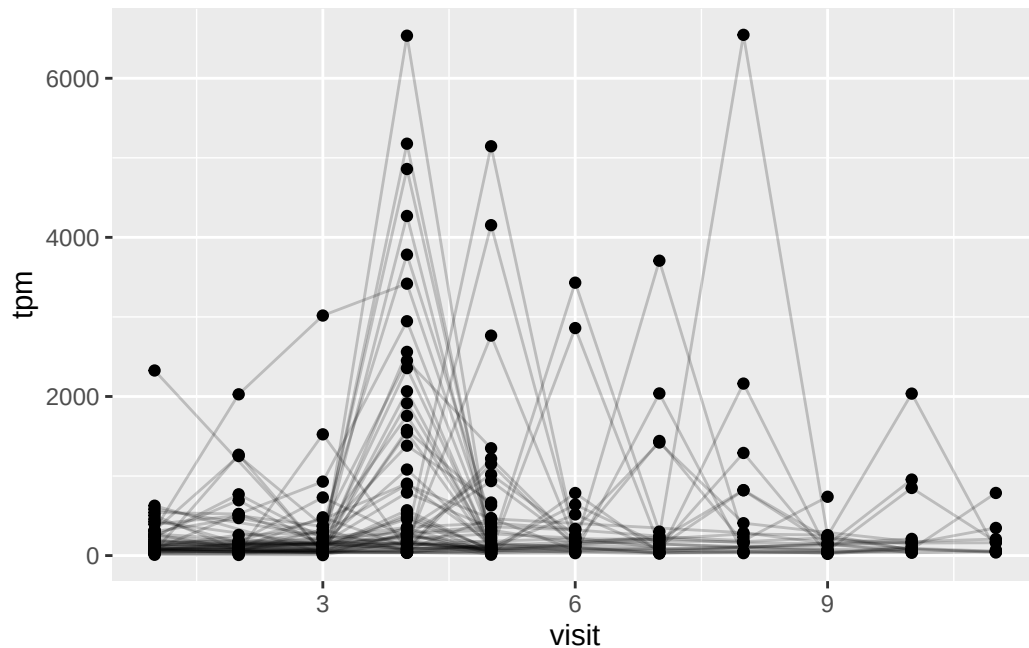
To facilitate further analysis we need to “join” the rna expression data with our metadata meta, which is itself a join of sample and specimen data. This will allow us to look at this genes TPM expression values over aP/wP status and at different visits (i.e. times):

```
#meta <- inner_join(specimen, subject)
ssrna <- inner_join(rna, meta)
```

Joining with `by = join\_by(specimen\_id)`

Q19. Make a plot of the time course of gene expression for IGHG1 gene.

```
ggplot(ssrna) +
  aes(x = visit, y = tpm, group=subject_id) +
  geom_point() +
  geom_line(alpha=0.2)
```



Q20. What do you notice about the expression of this gene (i.e. when is it at it's maximum level)?

The gene is at it's maximum level during visit 4 and visit 8.

Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

Yes, this pattern in time matches the trend of antibody titer data as the highest levels of antigen are during visits 4 and 8.